

Mohd Abbas Zaidi

Engineer, NLP Lab, Samsung Research HQ, Seoul

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🔗 scholar.google.com/citations?user=i_cSaKgAAAAJ

Education

- Jan 2023– **Master of Science, Georgia Tech (Online)**, Atlanta, USA.
Nov 2024 School of Computer Science (Artificial Intelligence, formerly Interactive Intelligence)
GPA – 4.0/4.0
- 2015–2019 **Bachelor of Technology, Indian Institute of Technology, Kanpur, India.**
Major – Electrical Engineering, Minors – Machine Learning & Applications | Cognitive Science.
GPA – 10.0/10.0 | **Institute Rank 1**
Thesis: Improved multimedia recovery in wireless MIMO systems – Advisor: Prof. Aditya Jagannatham

Work Experience

- 2019 – **Software Engineer | NLP Lab, Samsung Research HQ**, Seoul, South Korea.
Present **Generative AI Task Force**
- Trained Samsung's first in-house LLM, Gauss, for on-device deployment on mobile devices.
 - Improved training stability through principled initialization and scaling strategies for LLMs.
 - Implemented dynamic dataset adjustment to improve LLM training efficiency.
 - Downcycled LLMs by pruning them to commercially viable model sizes.
 - Applied knowledge distillation to boost on-device Gauss model performance.
- Question Answering | Machine Translation Group**
- Ranked first in the simultaneous translation challenge (low-latency regime) at ACL-IWSLT 2020.
 - Established a new end-to-end SOTA on the English–German MuST-C speech translation task.
 - Established a new SOTA for open-domain QA on the Natural Questions and Web Questions datasets.
- Recognition within Samsung:**
- Samsung Best Research Paper Award – Gold (2023); Bronze (2020, 2022, 2025).
 - Samsung Software Innovation Award – Winner (2021).
 - Samsung Foldable App Hackathon – Runner-up (2020).
- Patents:** Filed six patents related to language models, machine translation, and user experience.
Papers: Published at [ICML](#), [EMNLP](#), [ACL](#), [NAACL](#), [ACL-IWSLT](#), [ICASSP](#), and [Interspeech](#).
- Summer 2019 **Machine Learning & Development Internship, PocketFM**, Bengaluru, India.
Machine Learning
- Made search language-agnostic across Hindi, English, and Hinglish, both semantically and syntactically.
 - Built a recommendation system for the app using ALS matrix factorization on an RDD-backed database.
- Backend Development
- Deployed a centralized backend logging system (ELK stack) to automate error-log monitoring.
 - Deployed the Elastic-Logstash-Kibana stack to visualize backend health and provide burst-error alerts.
- Summer 2018 **Natural Language Processing Internship, Samsung Research HQ**, Seoul, South Korea.
Generating well-formed answers for AI assistants and chatbots.
- Topped the leaderboard for Microsoft's MS MARCO Intermediate QA challenge (June 2018).
 - Generated detailed, non-span-based answers using the ConZNet architecture.
 - Results featured on the Samsung [Newsroom](#) and other [AI/NLP](#) web forums.
- Summer 2017 **Research Internship, Indian Space Research Organisation–IITK Space Cell**, Kanpur, India.
Prof. Aditya Jagannatham | STTC + Total Variation-based Robust Multimedia Reconstruction.
- Incorporated the bounded-variation property of images into Space-Time Trellis Codes.
 - Designed a hierarchical joint decoder to jointly optimize the maximum-likelihood and total-variation metrics.
 - Published** in IEEE Access, Volume 8, April 2020.

Publications

* - Equal Contribution

- Jan 2025 **Sparse logit sampling: Accelerating knowledge distillation in LLMs**, [paper](#), ACL 2025 (Oral).
Anshumann*, Mohd Abbas Zaidi*, Akhil Kedia*, Jinwoo Ahn, Tahewak Kwon, Kangwook Lee, Haejun Lee, Joohyung Lee
- May 2024 **Transformers Get Stable: An End-to-End Signal Propagation Theory for Language Models**, [paper](#), ICML 2024.
Akhil Kedia*, Mohd Abbas Zaidi*, Sushil Khyalia*, Junggho Jung, Harshith Goka, Haejun Lee

- Oct 2022 **FiE: Building a Global Probability Space by Leveraging Early Fusion in Encoder for Open-Domain Question Answering**, [link](#), EMNLP 2022.
Akhil Kedia, Mohd Abbas Zaidi, Haejun Lee
- Oct 2021 **Cross Modal Decision Regularization for Simultaneous Speech Translation Systems**, [link](#), Inter-speech 2022 (Oral).
Mohd Abbas Zaidi*, Beomseok Lee*, Sangha Kim, Chanwoo Kim
- Sep 2021 **Infusing Future Information into Monotonic Attention Through Language Models**, [arxiv](#), NAACL 2022 .
Mohd Abbas Zaidi*, Sathish Indurthi*, Beomseok Lee, Nikhil Kumar, Sangha Kim
- Jan 2021 **Task Aware Multi-Task Learning for Speech to Text Tasks**, [link](#), ICASSP 2021.
Sathish Indurthi*, Mohd Abbas Zaidi*, Nikhil Kumar, Beomseok Lee, Hyojung Han, Seokchan Ahn, *et al.*
- May 2020 **Hierarchical Trellis Based Decoder for Total Variation Sequence Detection (TVSD) in Space-Time Trellis Coded (STTC) Wireless Image/Video Communication**, *IEEE Access Journal*, [link](#).
Ankit Kudeshia, Mohd Abbas Zaidi, Aditya Jagannatham, Chandra Prakash
- Jul 2020 **End-to-End Simultaneous Translation System for IWSLT2020 Using Modality Agnostic Meta-Learning**, *ACL-IWSLT 2020*, [link](#).
Houjeung Han, Mohd Abbas Zaidi, Sathish Indurthi, Nikhil Kumar, Beomseok Lee, Sangha Kim
- Jan 2021 **Faster Re-translation Using Non-Autoregressive Model For Simultaneous Neural Machine Translation**, *arxiv*, [link](#).
Hyojung Han, Sathish Indurthi, Mohd Abbas Zaidi, Nikhil Kumar, Beomseok Lee, *et al.*
- Jul 2020 **End-to-End Offline Speech Translation System for IWSLT 2020 using Modality Agnostic Meta-Learning**, *ACL-IWSLT 2020*, [link](#).
Nikhil Kumar, Beomseok Lee, Sathish Indurthi, Houjeung Han, Mohd Abbas Zaidi, Sangha Kim

Awards & Achievements

- Jun 2019 **Prof. Putcha Venketeswarlu Memorial Gold Medal**, *IIT Kanpur*, Highest GPA in the Class of 2019.
- Jun 2019 **General Proficiency Medal**, *IIT Kanpur*, Highest GPA in EE in the Class of 2019.
- Mar 2019 **Lalit Narain Das Memorial Scholarship**, *IIT Kanpur*, Highest GPA in EE during Senior Year.
- Feb 2019 **Smt Dharm Vati Garg Scholarship**, *IIT Kanpur*, Highest GPA overall during Senior Year.
- Mar 2018 **Anita Santoshi Mehra Foundation Scholarship**, *IIT Kanpur*, Highest GPA in EE during Junior Year.
- 2015-2018 **Academic Excellence Award**, *IIT Kanpur*, Awarded for three consecutive years.
- June 2015 **JEE**, 99.98 percentile among 1.3 million students across India in the university entrance exam..
- Jun 2017 **SURGE**, *IIT Kanpur*, Undergraduate Research Grant for Summer Internship.

Teaching/Volunteering

- Spring 2019 **Electrical Engineering Lab, IIT Kanpur**, *Teaching Assistant*, EE381A, *Prof. Baquer Mazhari*.
o Mentored student projects in the electronic circuit component of the EE lab course.
- Apr 2016 – **Fundamentals of Computing, IIT Kanpur**, *Academic Mentor*, ESC101A, Counselling Service.
- Apr 2017 o Conducted institute-wide lectures and doubt-clearing sessions for the introductory programming course.
- Feb 2017 – **Counselling Service, IIT Kanpur**, *Member, Core Team Operations*.
- Mar 2018 o Mentored students under academic probation and connected them with professional counsellors.